

## CHAPTER 5

### RESULTS AND EVALUATION

#### 5.1 Introduction

This chapter presents the results of the various approaches mentioned in the proposed model. The research methodology listed in Chapter 3 is executed with the hyperparameters listed in Chapter 4 and the results are summarized in this chapter. The experimental outcome of autoencoder based feature selection, unsupervised anomaly detection and clustering, and neural network classifiers are analyzed based on the evaluation metrics described in Chapter 3.

#### 5.2 AutoEncoder Based Feature Selection

The training and validation loss is plotted against the number of epochs for three variants of autoencoder based feature selection with the dimensionality of 25 and 16 on kdd\_raw, kdd\_prep1, and kdd\_prep2 datasets and shown in Figure 5.1, Figure 5.2, and Figure 5.3. The training and validation loss plots are smooth curves for AE and SAE feature selection techniques, However, the curves are irregular with peaks for Variational autoencoder feature selection techniques, The smooth curves are indicating a generalizable model. Also, it is observed that the autoencoder loss value is more for test datasets when compared to the corresponding training data and relatively high values of reconstruction error with VAE feature selection techniques. This could be attributed to the uneven distribution and more heterogenic nature of attack distribution in test datasets.

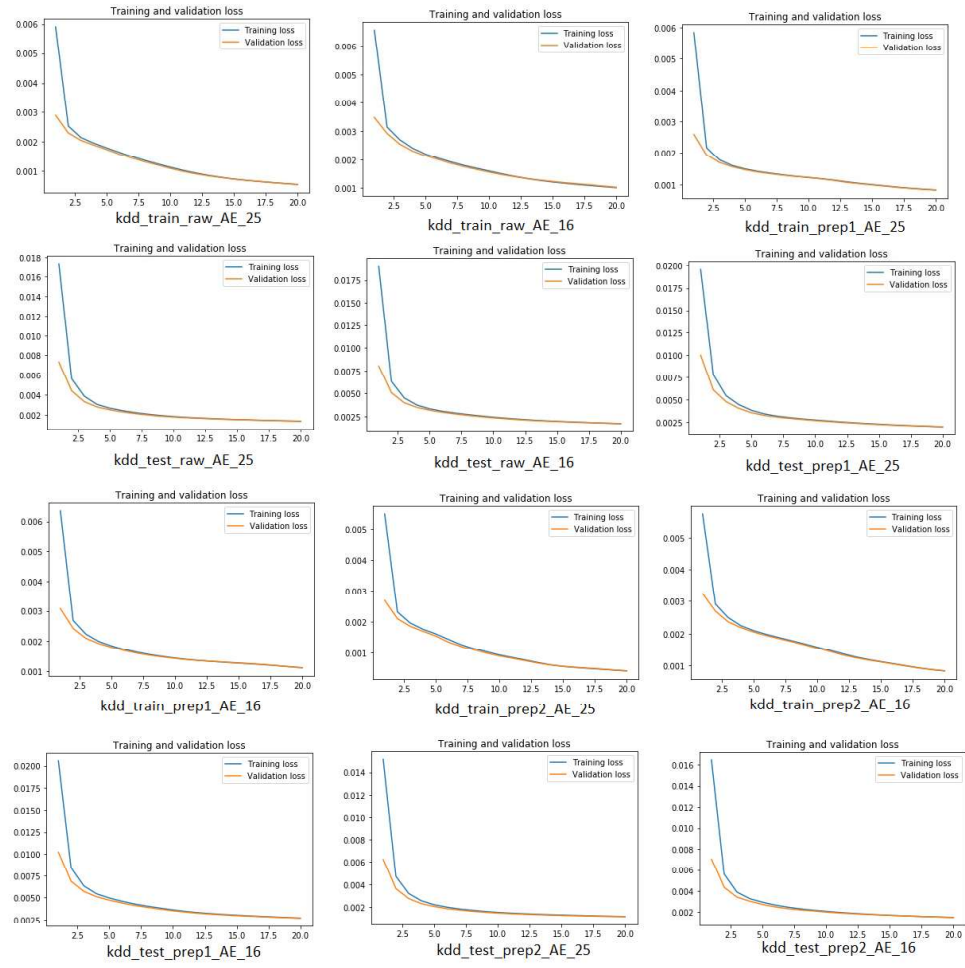


Figure 5.1 AE Training and Validation Loss

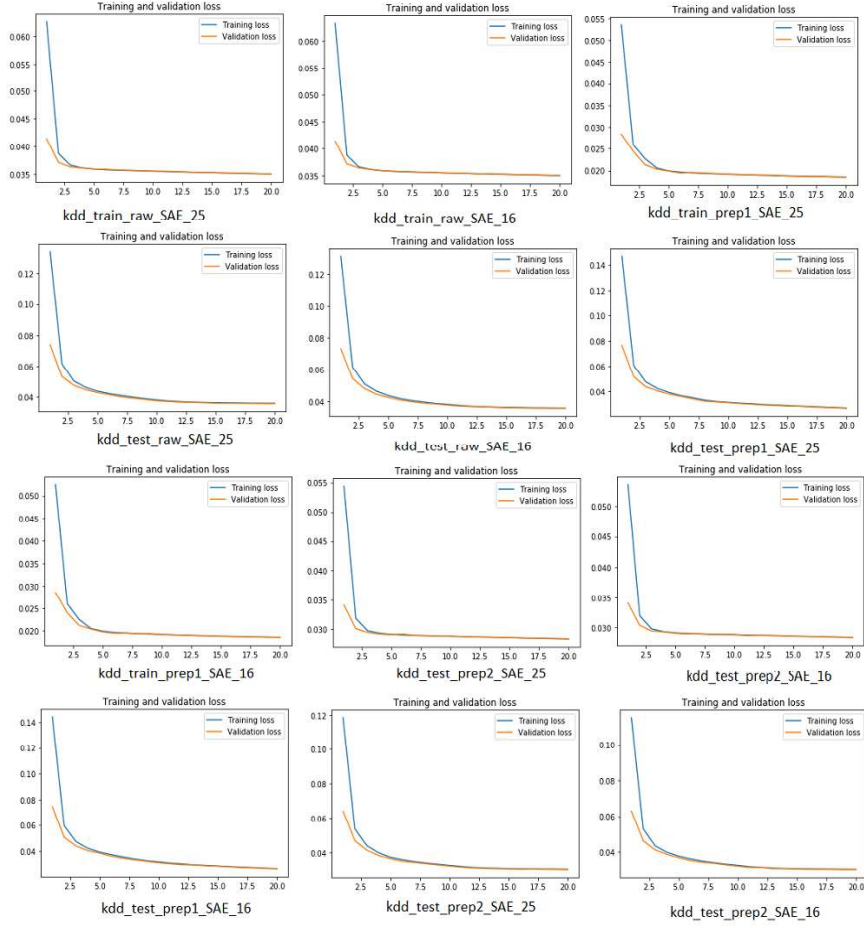


Figure 5.2 SAE Training and Validation Loss

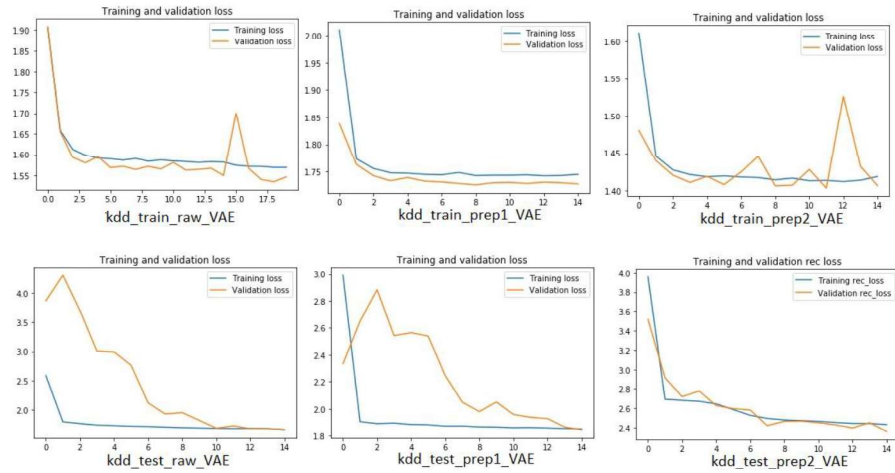


Figure 5.3 VAE Training and Validation Loss

### 5.3 Isolation Forest Metrics

The train and test datasets were passed to the Isolation Forest Classifier to categorize the traffic into Normal and attack. The evaluation metrics obtained for the three variants of data preparation are plotted in Figure 5.4 and Figure 5.5. The detailed confusion matrix parameters and clustering parameters are tabulated in Table 5.1.

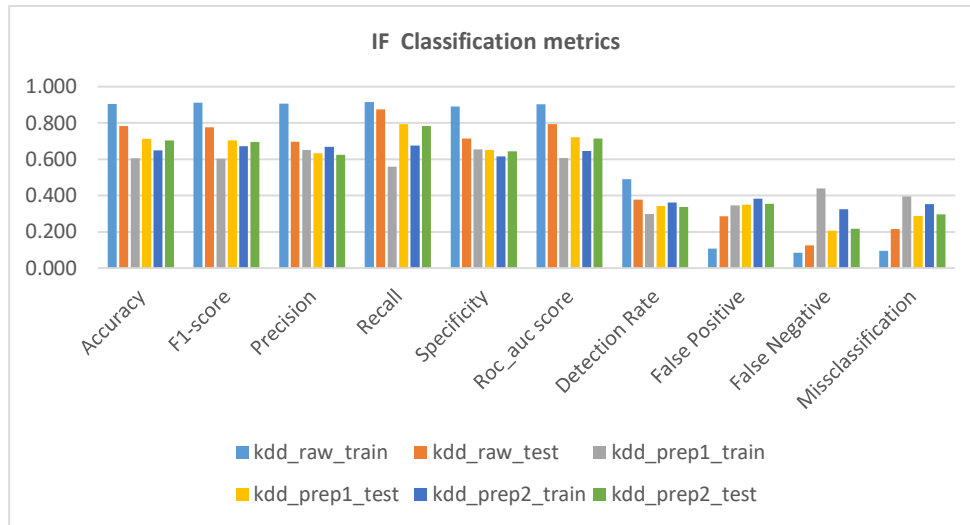


Figure 5.4 IF Evaluation Metrics

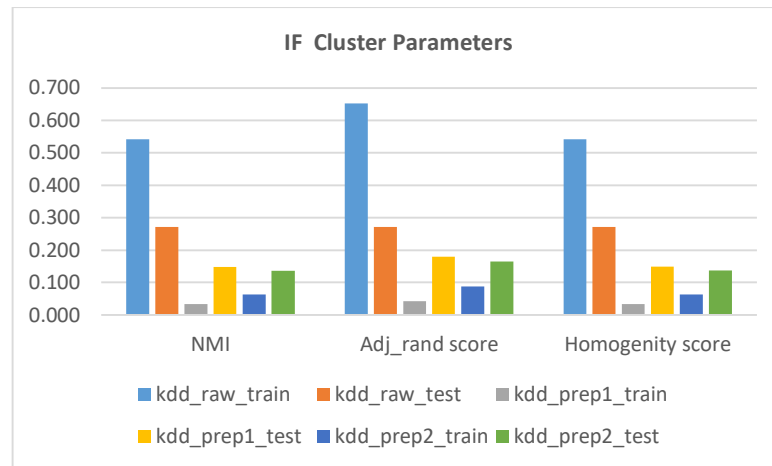


Figure 5.5 IF Cluster Parameters

Table 5.1 IF Evaluation Results

| ISOLATION FOREST    | kdd_prep2 |       | kdd_prep1 |       | kdd_raw      |              |
|---------------------|-----------|-------|-----------|-------|--------------|--------------|
|                     | train     | test  | train     | test  | train        | test         |
| Accuracy            | 0.648     | 0.704 | 0.604     | 0.712 | <b>0.904</b> | <b>0.783</b> |
| F1-score            | 0.673     | 0.695 | 0.602     | 0.704 | <b>0.911</b> | <b>0.776</b> |
| Precision           | 0.669     | 0.624 | 0.651     | 0.632 | <b>0.906</b> | <b>0.698</b> |
| Recall              | 0.676     | 0.783 | 0.560     | 0.793 | <b>0.915</b> | <b>0.875</b> |
| Specificity         | 0.617     | 0.644 | 0.655     | 0.651 | 0.891        | 0.713        |
| Roc_auc score       | 0.646     | 0.713 | 0.607     | 0.722 | 0.903        | 0.794        |
|                     |           |       |           |       |              |              |
| Detection Rate      | 0.361     | 0.337 | 0.299     | 0.342 | 0.489        | 0.377        |
| False Positive      | 0.383     | 0.356 | 0.345     | 0.349 | <b>0.109</b> | <b>0.287</b> |
| False Negative      | 0.324     | 0.217 | 0.440     | 0.207 | 0.085        | 0.125        |
| Missclassification  | 0.352     | 0.296 | 0.396     | 0.288 | <b>0.096</b> | <b>0.217</b> |
| Positive predictive | 0.669     | 0.624 | 0.651     | 0.632 | 0.906        | 0.698        |
| Negative Predictive | 0.624     | 0.797 | 0.564     | 0.806 | 0.902        | 0.883        |
|                     |           |       |           |       |              |              |
| NMI                 | 0.063     | 0.137 | 0.034     | 0.149 | 0.543        | 0.271        |
| Adj_rand score      | 0.088     | 0.166 | 0.043     | 0.180 | 0.653        | 0.271        |
| Homogeneity score   | 0.063     | 0.137 | 0.034     | 0.149 | 0.542        | 0.271        |
|                     |           |       |           |       |              |              |
| Run-time(sec)       | 9.419     | 2.437 | 9.682     | 2.109 | 9.697        | 2.152        |

The Isolation Forest Algorithm is giving optimal evaluation metrics for the kdd\_raw dataset with 90% accuracy, 0.90 ROC-AUC scores, and 0.91 F1score on the training dataset. As mentioned in the quantitative study of unsupervised algorithms by (Falcão et al., 2019) Isolation Forest performs well even with a dataset having high correlated variables... The confusion matrix for the test and train datasets are plotted in Figure 5.6.

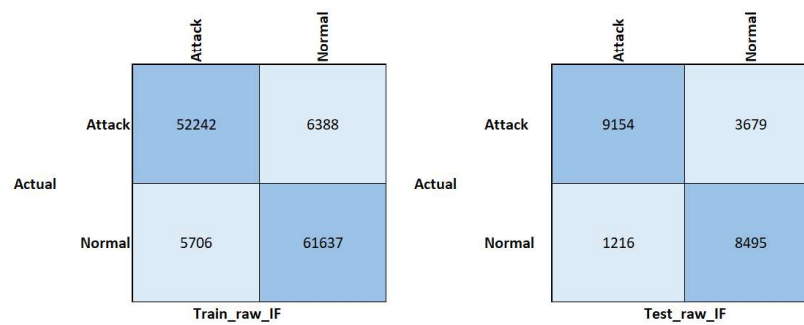


Figure 5.6 Confusion Matrix IF

## 5.4 K-means Evaluation Metrics

K-means Algorithm is implemented using the optimal hyperparameters on the feature selected dataset for unsupervised anomaly detection and attack clustering. The classification and cluster parameters for each set of the experimental dataset was plotted and analyzed.

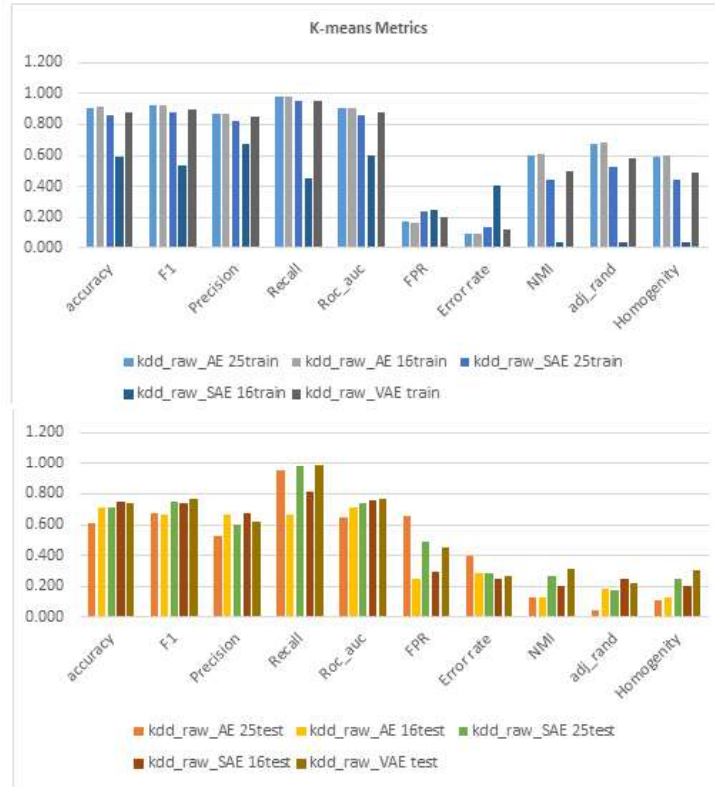


Figure 5.7 K-means kdd\_raw

Analyzing the K-means 2 cluster evaluation metrics of the train and test data, a similar pattern is observed among all the three data variants viz kdd\_raw, kdd\_prep1, and kdd\_prep2. AE\_25, AE\_16, and VAE are giving reasonable values with train evaluation while SAE\_16 performance on train K-means 2 cluster metrics is relatively poor. However, AE\_25 is not giving good results on the test dataset, while AE\_16, SAE\_25, and VAE are giving reasonable values on the test dataset. Considering both test and train metrics, a good trade-off between accuracy and FPR is shown by AE\_16.

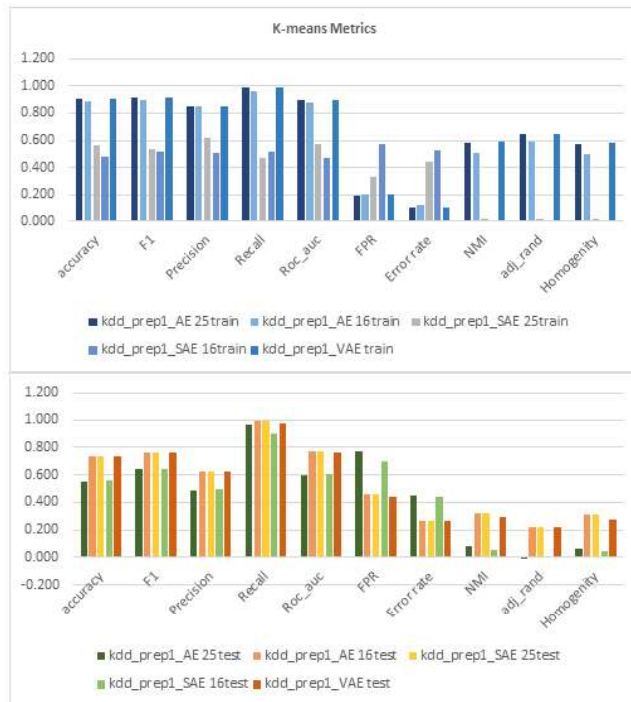


Figure 5.8 K-means kdd\_prep1

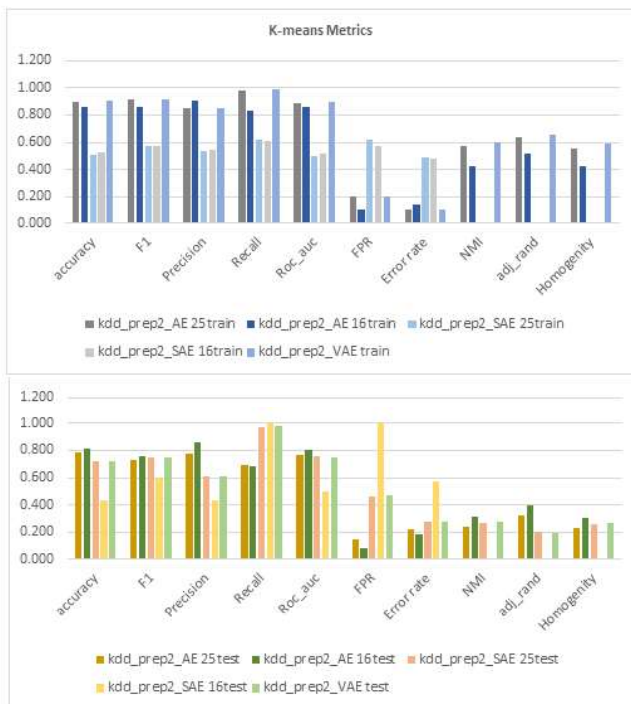


Figure 5.9 K-means kdd\_prep2

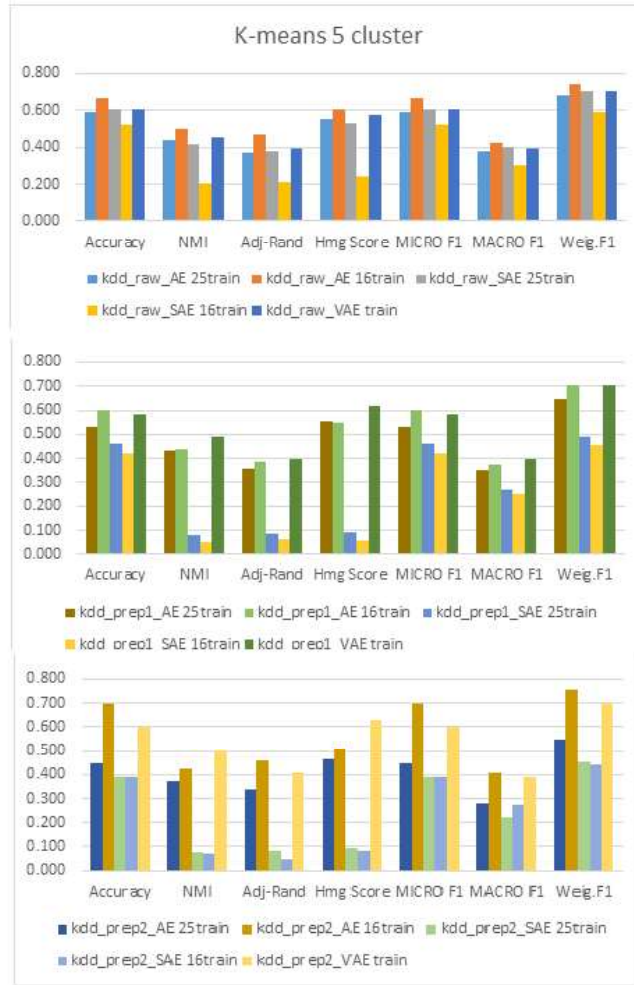


Figure 5.10 K-means 5 cluster

Analysis of K-means 5 cluster metrics among the three variants of train dataset viz kdd\_raw, kdd\_prep1, and kdd\_prep2, SAE\_16 IS the worst-performing feature selection technique. AE\_16 and VAE are giving comparatively higher metrics values for all three sets of data. The best performer is kdd\_prep2\_AE\_16 with the highest values on the accuracy, NMI and Weighted F1-score.



## 5.5 DNN classifier Metrics

The feature selected and K-means clustered datasets were classified using DNN classifiers aiming at performance improvement and the evaluation metrics for the data variants were plotted.

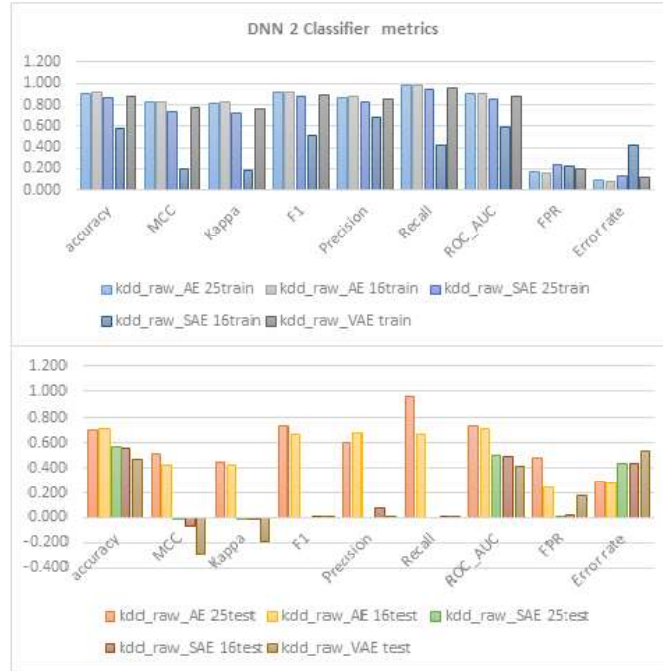


Figure 5.11 DNN 2 class metrics- kdd\_raw

Analyzing the DNN-2 classifier metrics, across the train and test datasets, SAE\_25 and SAE\_16 performed relatively poorly with very low and negative MCC and Kappa values, which is a clear indication of a below-average classifier. Similar to K-means cluster performance, DNN 2 classifier also performing well on the training dataset with AE\_25, AE\_16, and VAE. But AE\_25 is giving lower values on test data, while AE\_16 is showing reasonable evaluation performance with both train and test datasets. Comparing K-means 2 cluster and DNN-2 evaluation metrics, the best performer is kdd\_prep2\_AE\_16 with a significant improvement in test evaluation metrics.

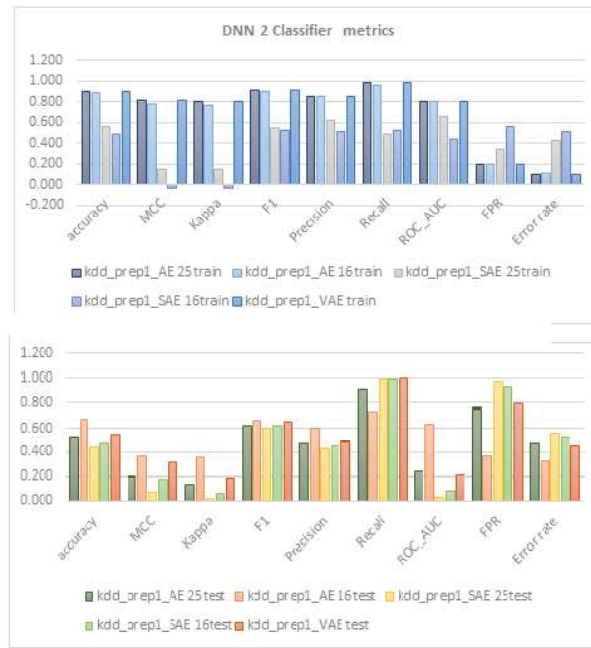


Figure 5.12 DNN 2 class metrics- kdd\_prep1

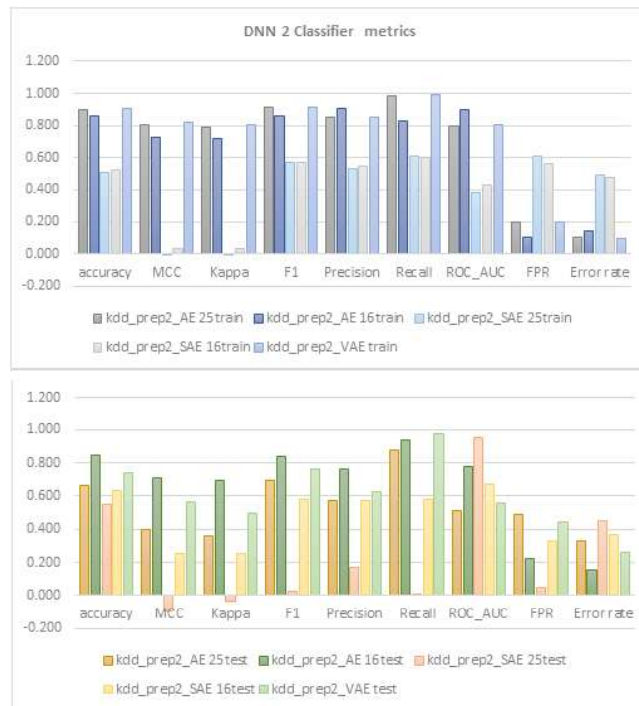


Figure 5.13 DNN 2 class metrics-kdd\_prep2

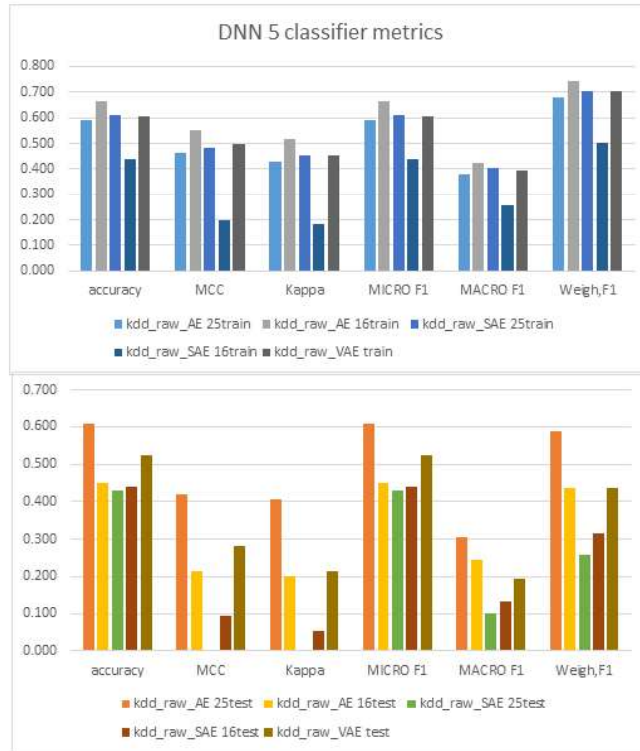


Figure 5.14 DNN 5 class metrics- kdd\_raw

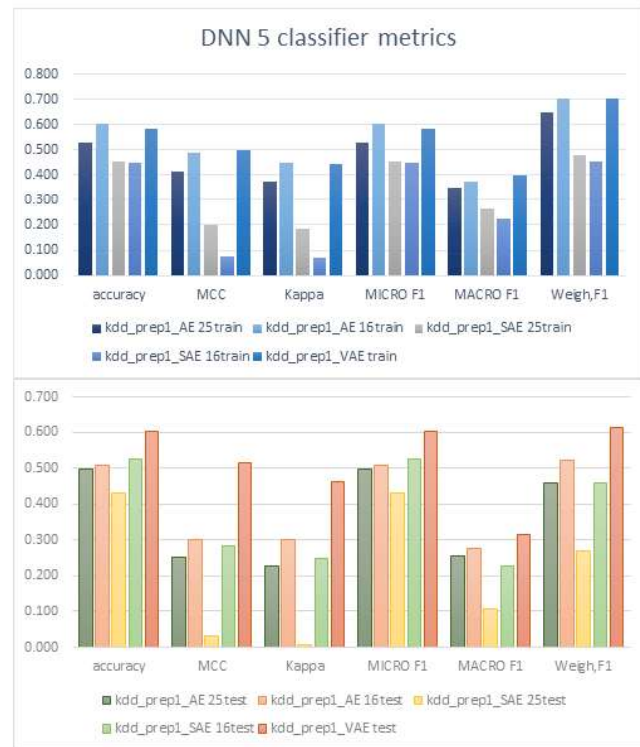


Figure 5.15 DNN 5 class metrics- kdd\_prep1



Figure 5.16 DNN 5 class metrics-kdd\_prep2

The multiclassification metrics of the DNN-5 classifier on feature selected and K-means 5 cluster datasets were evaluated and plotted. Analysis of the DNN-5 classifier metrics across datasets revealed that SAE\_25 and SAE\_16 are poor models with very low MCC and Kappa scores. AE\_25, AE\_16 and VAE are giving an overall better model performance. Among the data preparation variants, the overall model parameters are best for kdd\_prep2\_VAE

## 5.6 Time of Execution

The total run time of the model for training datasets includes the execution time of autoencoder based feature selection, k-means clustering, and the DNN classification. For the test datasets, the total run time consists of feature selection and DNN classification. The runtime calculated for data variants that showed optimal performance for DNN 2 class and DNN 5 class are plotted and shown in Figure 5.17 and Figure 5.18,



Figure 5.17 Run time- 2 class

The highest computational time for binary classification among Feature Selection techniques is with Variational Autoencoders, whereas kdd\_prep1 datasets have more computational time owing to their increased original dimensionality. VAE FS consumes more time for sampling and calculating the mean and variance of the samples. The lowest computational time on test data is shown by AE\_16 FS.

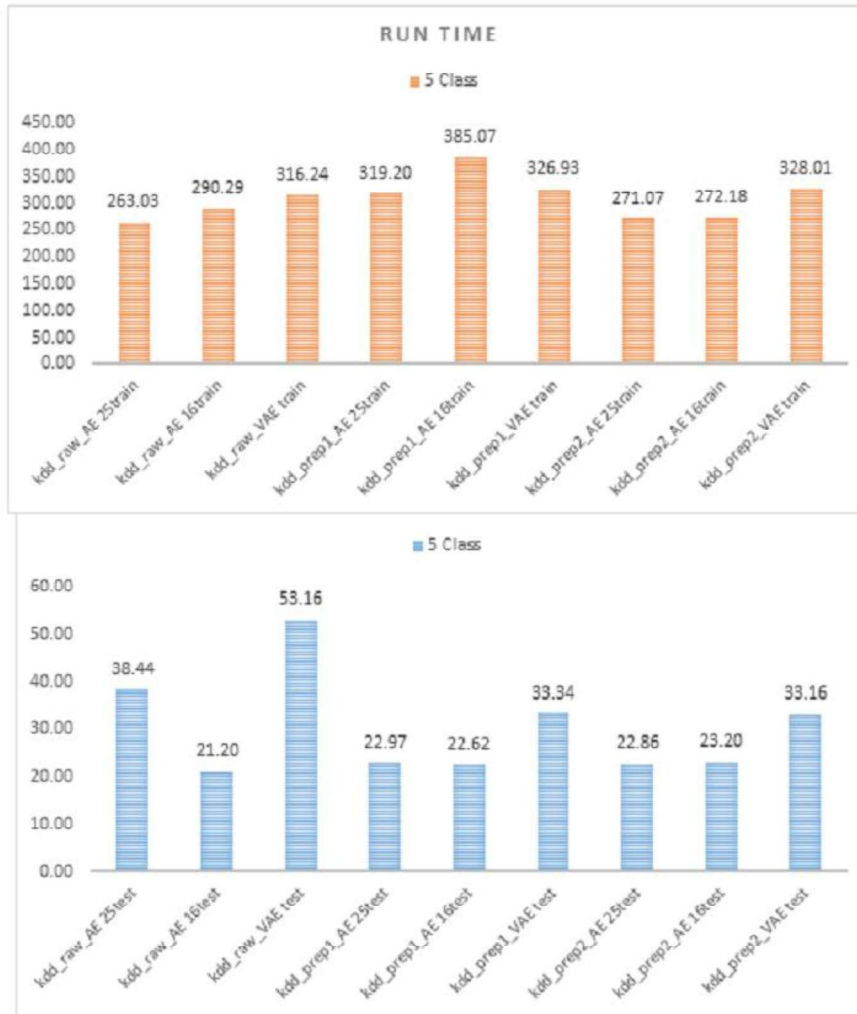


Figure 5.18 Run time- 5 class

The highest computational time for 5 class classification among Feature Selection techniques is with Variational Autoencoders, whereas kdd\_prep1 datasets have more computational time owing to their increased original dimensionality. VAE FS consumes more time for sampling and calculating the mean and variance of the samples. The lowest computational time on test data is shown by AE\_16 FS.

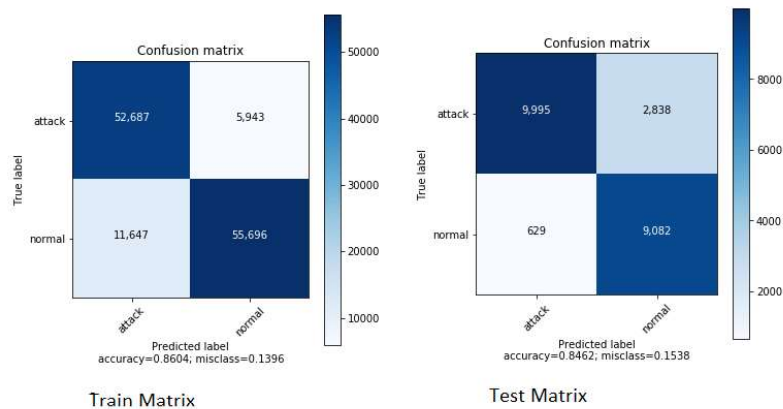
## 5.7 Findings

### Anomaly Detection

The evaluation metrics of IF, K-means 2 cluster, and DNN-2 classifier is summarized in Table 5.2 for analyzing the performance improvement upon unsupervised method with Neural network classifier. The optimal performance metrics for anomaly detection were given by the kdd\_prep2\_AE\_16\_KM dataset with a train and test accuracy of 0.86 and 0.84 respectively. The time of execution of the test dataset comes around 23.17 sec. The kdd\_prep2\_AE\_16 has achieved 20% performance improvement in recall and accuracy with AE FS, K-means, and DNN classification over IF Anomaly Detection.

However, the proposed model failed to classify novel attacks. The 293 unknown attacks were classified as Normal traffic with nil accuracy. A comparison of performance evaluation of proposed model on NSL KDD test Dataset with other Anomaly Detection Method as in Table 5.4 affirms that the proposed hybrid model was able to achieve test metrics on par with other semi-supervised and supervised techniques. The model achieved a detection rate or recall higher than other autoencoder based semi-supervised approaches and a comparable F1 score.

Table 5.2 Confusion Matrix



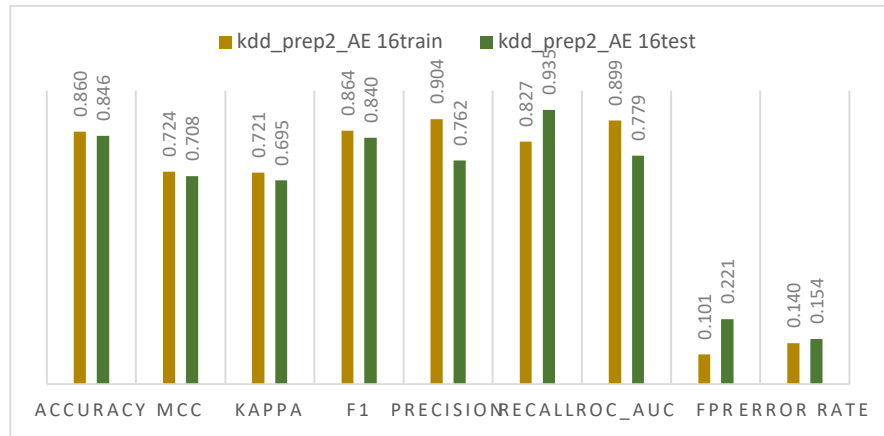


Figure 5.19 Evaluation Metrics

Table 5.3 Attackwise Performance

| pred    | Attack | Normal |
|---------|--------|--------|
| Actual  |        |        |
| DoS     | 6831   | 336    |
| Normal  | 629    | 9082   |
| Probe   | 2200   | 221    |
| R2L     | 949    | 1936   |
| U2R     | 15     | 52     |
| unknown | 0      | 293    |

Table 5.4 Comparison of NSL-KDD Test Anomaly Detection

| Author                  | Type                | Accuracy     | Recall       | F1-score     |
|-------------------------|---------------------|--------------|--------------|--------------|
| (Aygun and Yavuz, 2017) | Semi supervised     | 0.887        |              |              |
| (Falcão et al., 2019)   | Unsupervised        | 0.753        |              |              |
| (Pu et al., 2019)       | Unsupervised        |              | 0.890        |              |
| (Pérez et al., 2020)    | Unsupervised        | 0.850        |              | 0.870        |
| (Su et al., 2020)       | Supervised          | 0.901        | 0.985        | 0.904        |
| (Dlamini et al., 2019)  | Semi supervised     |              | 0.880        | 0.880        |
| <b>Proposed model</b>   | <b>Unsupervised</b> | <b>0.846</b> | <b>0.935</b> | <b>0.840</b> |



Table 5.5 Anomaly Detection Metrics Summary

| Algorithm      | Isolation Forest |       | K-means 2  |              |          | NN-2 Classifier |               |             |
|----------------|------------------|-------|------------|--------------|----------|-----------------|---------------|-------------|
| kdd_raw        | train            | test  | AE 25 test | AE 16 test   | VAE test | AE 25_KM test   | AE 16_KM test | VAE_KM test |
| Accuracy       | 0.904            | 0.783 | 0.607      | 0.714        | 0.736    | 0.707           | 0.719         | 0.468       |
| F1-score       | 0.911            | 0.776 | 0.676      | 0.667        | 0.764    | 0.738           | 0.671         | 0.002       |
| Precision      | 0.906            | 0.698 | 0.524      | 0.668        | 0.622    | 0.600           | 0.677         | 0.006       |
| Recall         | 0.915            | 0.875 | 0.955      | 0.666        | 0.990    | 0.960           | 0.665         | 0.002       |
| Specificity    | 0.891            | 0.713 | 0.343      | 0.750        | 0.545    | 0.516           | 0.760         | 0.821       |
| ROC_AUC score  | 0.903            | 0.794 | 0.649      | 0.708        | 0.767    | 0.738           | 0.713         | 0.411       |
| Detection Rate | 0.489            | 0.377 | 0.411      | 0.287        | 0.426    | 0.413           | 0.287         | 0.001       |
| False Positive | 0.109            | 0.287 | 0.657      | 0.250        | 0.455    | 0.484           | 0.240         | 0.179       |
| kdd_prep       | train            | test  | AE 25 test | AE 16 test   | VAE test | AE 25_KM test   | AE 16_KM test | VAE_KM test |
| Accuracy       | 0.604            | 0.712 | 0.547      | 0.736        | 0.737    | 0.529           | 0.678         | 0.549       |
| F1-score       | 0.602            | 0.704 | 0.647      | 0.765        | 0.761    | 0.624           | 0.665         | 0.656       |
| Precision      | 0.651            | 0.632 | 0.487      | 0.621        | 0.625    | 0.475           | 0.603         | 0.489       |
| Recall         | 0.560            | 0.793 | 0.963      | 0.994        | 0.974    | 0.908           | 0.743         | 1.000       |
| Specificity    | 0.655            | 0.651 | 0.233      | 0.541        | 0.557    | 0.575           | 0.686         | 0.604       |
| Roc_auc score  | 0.607            | 0.722 | 0.598      | 0.768        | 0.766    | 0.242           | 0.629         | 0.208       |
| Detection Rate | 0.299            | 0.342 | 0.415      | 0.428        | 0.419    | 0.391           | 0.320         | 0.431       |
| False Positive | 0.345            | 0.349 | 0.767      | 0.459        | 0.443    | 0.758           | 0.371         | 0.792       |
| kdd_prep2      | train            | test  | AE 25 test | AE 16 test   | VAE test | AE 25_KM test   | AE 16_KM test | VAE_KM test |
| Accuracy       | 0.648            | 0.704 | 0.782      | <b>0.816</b> | 0.721    | 0.667           | <b>0.846</b>  | 0.738       |
| F1-score       | 0.673            | 0.695 | 0.732      | <b>0.762</b> | 0.751    | 0.693           | <b>0.840</b>  | 0.763       |
| Precision      | 0.669            | 0.624 | 0.779      | <b>0.859</b> | 0.610    | 0.575           | <b>0.762</b>  | 0.625       |
| Recall         | 0.676            | 0.783 | 0.690      | <b>0.685</b> | 0.979    | 0.873           | <b>0.935</b>  | 0.977       |
| Specificity    | 0.617            | 0.644 | 0.852      | <b>0.915</b> | 0.526    | 0.692           | <b>0.857</b>  | 0.767       |
| Roc_auc score  | 0.646            | 0.713 | 0.771      | <b>0.800</b> | 0.752    | 0.511           | <b>0.779</b>  | 0.557       |
| Detection Rate | 0.361            | 0.337 | 0.297      | <b>0.295</b> | 0.422    | 0.376           | <b>0.403</b>  | 0.421       |
| False Positive | 0.383            | 0.356 | 0.148      | <b>0.085</b> | 0.474    | 0.489           | <b>0.221</b>  | 0.443       |

### Attack Classification

The five-class model achieved average results in terms of accuracy and F1 score in a complete self-learned manner and there is scope for performance enhancement. The variational autoencoder feature selection surpassed AE and SAE in multi-class classification. VAE could better learn the distinguishing feature sets for different attack types due to its generative capability. However, the model failed to identify the unknown attacks.

The optimal multi-class performance is given by kdd\_prep2\_VAE data. With accuracy and weighted F1 score of (0.60,0.7 )and (0.591, 0.65) for train and test data respectively with a test execution time of 33 seconds. The DNN 5 overall model metrics for train and test data are plotted in Figure 5.20.

Comparing the individual attack metrics with similar works done as tabulated in Table 5.7, it is observed that our model could achieve lower values of false alarm rate on different attack types, but at the cost of the detection rate for Normal and DoS. The optimal attack wise metrics for kdd\_prep2\_VAE is given in Table 5.6, Figure 5.21, and Figure 5.22.

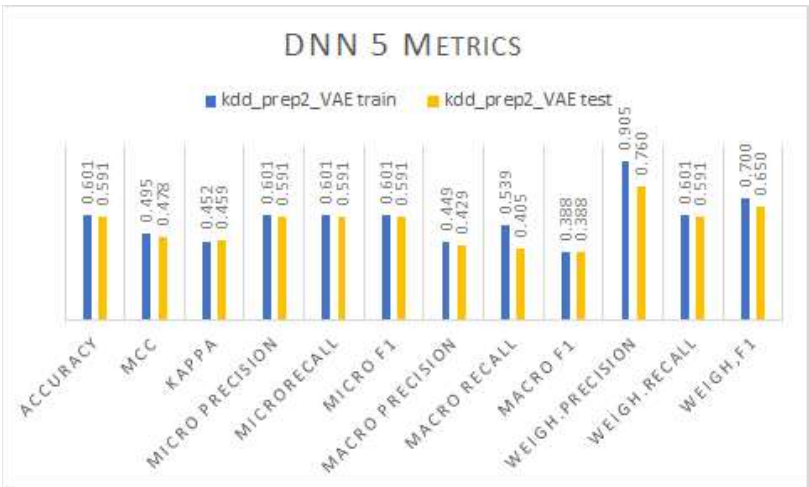


Figure 5.20 DNN 5 class model metrics

Table 5.6 Muti class Confusion Matrix

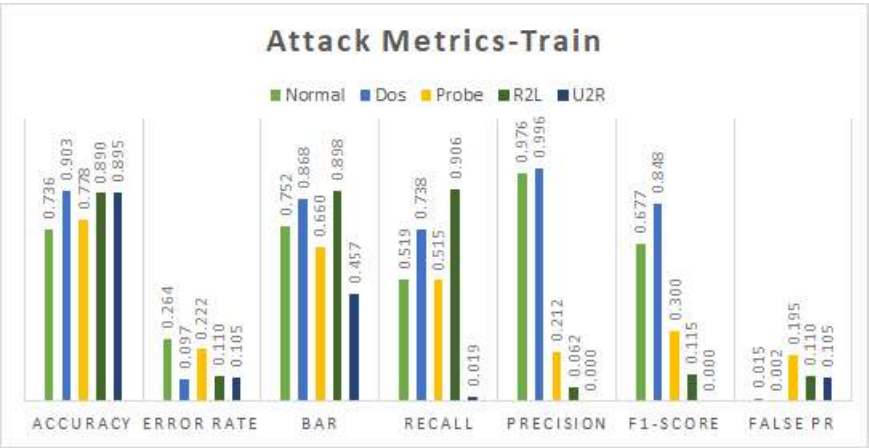
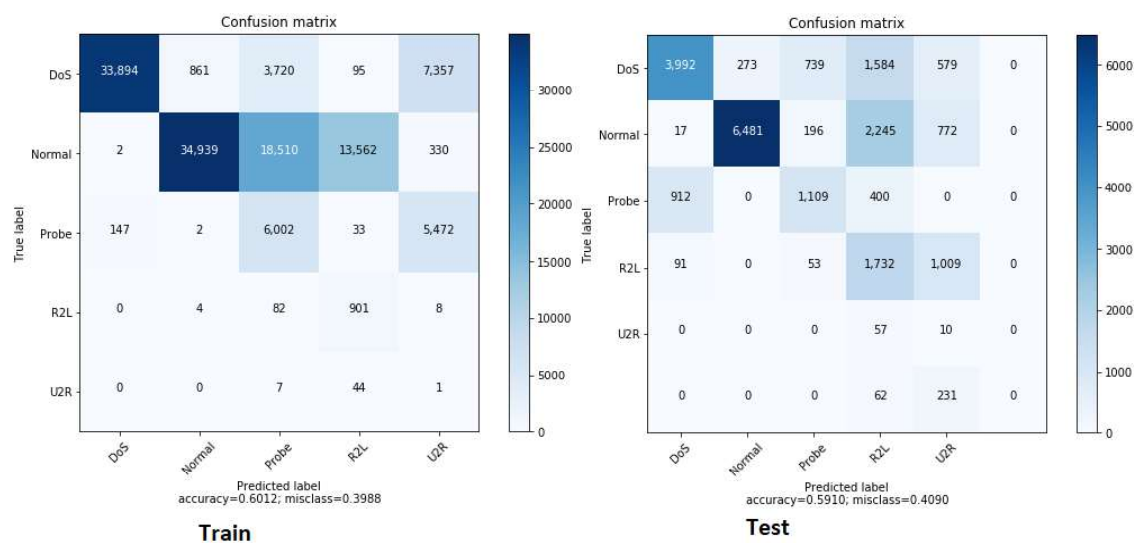


Figure 5.21 Attack Metrics-Train

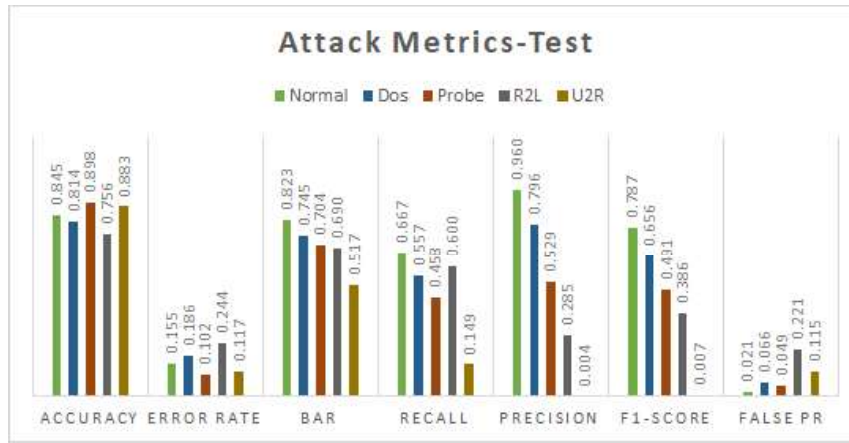


Figure 5.22 Attack Metrics-Test

Table 5.7 Comparison of Five-class Attack metrics

| Sno | Paper              | Method        | Algorithm      | Attack Type | Detection Rate | False Alarm Rate |
|-----|--------------------|---------------|----------------|-------------|----------------|------------------|
| 1   | (Pu et al., 2019)  | Unsuper vised | SSC-OCSVM      | Normal      | 89             | 8                |
|     |                    |               |                | Dos         | 85             | 7.8              |
|     |                    |               |                | Probe       | 99             | 6.6              |
|     |                    |               |                | R2L         | 52             | 8.95             |
|     |                    |               |                | U2R         | 90             | 9.05             |
| 2   | (Su et al., 2020)  | Deep Learning | BLSTM          | Normal      | 97.5           | 25.7             |
|     |                    |               |                | Dos         | 87.55          | 1.52             |
|     |                    |               |                | Probe       | 85.76          | 1.15             |
|     |                    |               |                | R2L         | 44.25          | 0.91             |
|     |                    |               |                | U2R         | 20.95          | 0.09             |
| 3   | (Yin et al., 2017) | Deep Learning | RNN            | Normal      | 96             | 10.01            |
|     |                    |               |                | Dos         | 83.49          | 2.06             |
|     |                    |               |                | Probe       | 83.4           | 2.16             |
|     |                    |               |                | R2L         | 24.69          | 0.8              |
|     |                    |               |                | U2R         | 11.5           | 0.07             |
| 4   | Proposed Model     | Hybrid        | AE-K-means-DNN | Normal      | 66.74          | 2.13             |
|     |                    |               |                | Dos         | 55.70          | 6.63             |
|     |                    |               |                | Probe       | 45.81          | 4.91             |
|     |                    |               |                | R2L         | 60.03          | 22.12            |
|     |                    |               |                | U2R         | 14.93          | 11.53            |

In 2 class and 5 class classification, kdd\_prep2 was giving better performance compared to kdd\_raw and kdd\_prep1. Thus a data prepared with high absolute correlation dropped and only the categorical variables dummy encoded and nil variant values neglected could be considered as a good candidate for anomaly detection. Dummy encoding of ordinal values brought out only the overhead of parameters and computational time, not a performance enhancement.

The moderate performance accomplished by sparse autoencoder based experiments may be suspected to the L1 regularizer used hidden layers. An L1 regularizer was not sufficient to deal with the nonlinearity associated with the dataset attributes. Whereas AE and VAE better performed with the nonlinear activation functions ‘elu’ and ‘tanh’.Also, the non-linear activation function used in DNN layers thrived to improve the model performance.

Upon inspecting the confusion metrics of training data, the Normal traffic is being misclassified majorly as R2L and U2R attack types whereas DoS traffics are being misinterpreted as Probe. To some extent, similar interpretation is visible in test data also. This affirms the insights derived from the bivariate analysis in Section 4.4. The performance enhancement of the unsupervised multi-attack classification may be achieved with in-depth feature selection or clustering techniques.

## **5.8 Summary**

This chapter presented the experimental findings and analysis of the different steps explained in the project pipeline. Firstly, the autoencoder training and validation loss functions were analyzed, Then the Isolation Forest metrics were evaluated for the three data variants. Next K-means 2 cluster and K-means 5 cluster metrics for the AE, SAE, and VAE feature selected data variants were analyzed. Then the DNN 2 and DNN-5 classifier metrics were analyzed. The IF, K-means, and DNN metrics for the 2 classes were compared for understanding the performance enhancements. The execution time for 2 Class and 5 class classification was calculated and compared. Finally, the proposed model statistics were compared with similar literature works and the findings were summarized.

## CHAPTER 6

### CONCLUSIONS AND RECOMMENDATIONS

#### 6.1 Introduction

The enormous growth of web-based application demands robust cybersecurity measures and performance enhancement of Network Intrusion Detection systems. This thesis is an attempt toward the development of an unsupervised Anomaly Detection System blending machine and deep learning techniques. In this section, the thesis work is discussed and the contribution and future recommendations are presented

#### 6.2 Discussions and Conclusion

A thorough analysis of the cyber attack statistics published emphasizes, the significance of Network Intrusion Detection Systems from a global economic and business perspective. Around 60 papers on outlier and intrusion detection techniques from 2004 to 2020 have been reviewed and found that some of the detection techniques, such as machine learning, deep learning play a vital role in implementing cost-effective IDS.

A detailed literature review is provided on these techniques with their background, applications, and limitations in the area of intrusion detection. Based on the thorough study of around 30 papers on ML and DL based network anomaly detection, a performance improvement is envisaged in a completely unsupervised method of anomaly detection and attack classification and also detecting zero-day attacks.

With that aim, an intrusion detection model was proposed involving autoencoder-based feature selection, self-learned anomaly detection, and attack clustering using unsupervised machine learning algorithms and performance improvement with Deep Neural Networks. The model would then be evaluated using the benchmark NSL-KDD dataset. In this work, three feature selection methods, Autoencoder, Sparse Autoencoder, and Variational Autoencoder were being considered for dimensionality reduction. The unsupervised algorithms Isolation Forest and K-means were used for anomaly detection. Then a DNN classifier would be trained and validated on the self-learned traffic clusters and performance evaluated on the test dataset.

The experiments were done on a GPU enabled cloud platform and coding was implemented in Python using Keras and Tensorflow as the backend. The data was preprocessed with necessary data cleansing, outlier treatment, attack categorization, categorical encoding, and normalization after exploratory data analysis. The algorithms were finely tuned and validated for selecting optimal hyperparameters. Then the model was evaluated on test data set using sklearn.metrics library. The results were visualized using seaborn and matplotlib libraries.

The results of the experiment showed that autoencoders trained in an unsupervised manner can be used for dimensionality reduction and feature selection in anomaly detection. Even though sparse autoencoder performed moderately, autoencoders and VAE accomplished better results. Anomaly Detection with DNN classifier upon K-means clustering has achieved 20% improvement in accuracy and recall when compared to the conventional Isolation Forest classification.

Even though the attack categorization and response to novel attack results were average, the model was able to achieve lower false alarm rates on different attack types with a relatively lesser computation time. The attack categorization and novelty detection capability have to be enhanced by further fine-tuning of parameters and deploying other clustering techniques. The bivariate analysis along with the final attack clusters suggests that a feature ensembling will help in distinguishing normal and attack types more accurately.

The results have also shown that hybrid approaches combining ML and DL techniques achieved satisfactory outputs for anomaly detection, and hence, these methods can be exploited in future research. This thesis focused on anomaly detection on static data only, which could be extended in the future for real-time data.

### **6.3 Contribution**

The thesis aimed to predict the anomalies from unlabeled network traffic with the highest accuracy and our results is promising enough and on par with the state-of-the-art supervised classifiers performance. Moreover, the classification time is relatively less which makes it recommendable for real-time anomaly detection. Thus our model could be considered as a cost-effective and time-efficient approach for Network Intrusion Detection.

#### **6.4 Future recommendations**

In future works, the proposed methods shall be elaborated with other feature selection approaches such as Stacked autoencoders and non-symmetric autoencoders to achieve better anomaly detection. Furthermore exploring semi-supervised techniques by extracting Normal network traffic and then performing attack clustering shall be suggested for achieving higher accuracy especially for Probe, R2L, and U2R attack categories.